

1. Critical issues

1.1 Dashboard Quality section map is still using the old I–O layout

The current section registry still defines:

SECTION I - FRAMEWORK HEALTH CHECK

SECTION J - SIGNOFF

SECTION K - SUMMARY

SECTION L - PERFORMANCE SUMMARY

SECTION M - MASTER LIST HEALTH CHECK

SECTION N - CARE PLAN SYNC DIAGNOSTICS

SECTION O - WORKFLOW & SYNCHRONIZATION VERIFICATION

That does not match the requested new layout where H–N should be:

SECTION H - FRAMEWORK HEALTH CHECK

SECTION I - PERFORMANCE SUMMARY

SECTION J - MASTER LIST HEALTH CHECK

SECTION K - CARE PLAN SYNC DIAGNOSTICS

SECTION L - WORKFLOW & SYNCHRONIZATION VERIFICATION

SECTION M - SUMMARY

SECTION N - SIGNOFF

The current incorrect registry is in RFF_DASHBOARD_QUALITY_SECTIONS.

The current runDashboardQualityFull() execution order is also still old and writes Signoff/Summary before Performance/Master/Care Plan/Workflow, which means Summary/Signoff can summarize stale or missing downstream sections.

1.2 Signoff still includes removed / relocated checks

The PDF reports show SECTION J - SIGNOFF still containing:

Intentional Validation Failure - MISSING

Monthly Change Subheaders - MISSING

That is consistent with the source: `appendCombinedDashboardSignOffRows_()` still pushes both removed/relocated rows into signoff.

Problems:

Intentional Validation Failure was removed and should not appear in Signoff.

Monthly Change Subheaders should be under Section F - Template Structure & Validation, not Signoff.

Signoff should move to Section N, not remain Section J.

1.3 Dashboard Audit is still part of Summary/Signoff logic even though it was removed
The source still computes dashboard audit status for Signoff.

The source also still includes Dashboard Audit as a Summary row mapped to Dashboard Quality Report Section H.

That conflicts with your instruction:

SECTION K - SUMMARY Dashboard Audit - Removed

Under the new mapping, Section H is Framework Health Check, not Dashboard Audit.

1.4 Summary rows still reference old sections L–O
`buildFrameworkSummaryRows_()` still reports:

Performance Summary → Section L

Master List Health Check → Section M

Care Plan Sync Diagnostics → Section N

Workflow & Synchronization Verification → Section O

Those references must shift to I–L, and Summary itself must become Section M.

1.5 Section update functions hard-code old Summary/Signoff section titles
`updateDashboardQualitySignoffSection_()` still writes SECTION J - SIGNOFF.

`updateDashboardQualitySummarySection_()` still writes SECTION K - SUMMARY.

These need to become:

SECTION M - SUMMARY

SECTION N - SIGNOFF

1.6 Dashboard Quality report headers still include old per-section header assumptions
The default header switch still expects:

H = generic audit-style Area / Item / Status

I = Health

J = Signoff

K = Summary

L = Performance

M = Master List

N = Care Plan

O = Workflow

That must be shifted so:

H = Health

I = Performance

J = Master List

K = Care Plan

L = Workflow

M = Summary

N = Signoff

Current header routing is here.

1.7 Legacy Dashboard Quality / framework-test storage still references removed tests
The code still has storage writers for:

Test 7

Test 9

Test 10 Dashboard Audit

These are still able to save rows under removed or relocated keys.

Even if the runtime wrappers are removed later, these storage paths can keep stale entries alive in Summary/Signoff unless cleaned.

1.8 applyFinalRowHeightLock_ is not defined caused Create Disenrolled failure in the reports
The Google Apps Script Executions PDF shows createDisenrolledList failed with:

ReferenceError: applyFinalRowHeightLock_ is not defined

The Framework Timing Report also records this as a CRITICAL issue. This matches the user-reported blocker. In the source lineage, the fast v1.4.26.1 archive has a defined applyFinalRowHeightLock_() and a wrapper applyFinalRowHeightLockForSheetType_().

In the current script, the equivalent active helper is lockFinalOutputRowHeights_(), which applies dashboard/global row-height settings.

2. Performance bottlenecks

2.1 Dashboard Quality Workflow is the largest bottleneck

From Report/v1.5.52 - Framework Timing Report.pdf:

First Dashboard Quality Workflow run: 357.1 sec

Second Dashboard Quality Workflow run: 296.999 sec

This is the most severe bottleneck. The current implementation calls each section through runDashboardQualitySectionIfDue_(), and those child functions frequently run their own timed/report-writing paths.

Likely causes:

nested timing/report generation

multiple individual section writes

repeated style passes

Summary/Signoff running before dependent sections

Framework Timing report being rewritten inside Dashboard Quality Workflow

2.2 Template refresh first-build is slow but below its 240-sec benchmark
From the Framework Timing PDF:

Create / Refresh All Templates: 210.257 sec

Slow steps called out:

Apply column widths: Template - Raw Data — 11.188 sec

Set template column count from dashboard headers: Template - Disenrolled Exclusion (64 cols)
— 10.576 sec

The current template build still applies broad sheet-level formatting, then title rows, header row, data row base style, column widths, date/number formats, hidden settings, filters, subheaders, and row-height skip markers.

Column width application is grouped into runs, which is good, but Raw Data likely has enough varied column widths to fragment runs and increase calls.

2.3 Raw Data output creation is critical
From the Framework Timing PDF:

Format Raw Data: 169.091 sec

Critical step:

Create Raw Data output from template skeleton and paste values: 150.151 sec

This points to template-copy/output-write cost, not just formatting. The architecture should ensure Raw Data output does not perform any full-sheet formatting or broad resizing after the copy.

2.4 Process Demo P is a bottleneck
From the Framework Timing PDF:

Process Demo P: 83.505 sec

Bottleneck step:

Copy active Raw Data rows to Demo P template: 39.359 sec

Slow step:

Single write processed Demo P data: 19.939 sec

This is probably data volume + template-copy/write cost. It should be optimized with chunked writes only when necessary and no redundant format copies after the template copy.

2.5 Show/hide templates/system sheets are expensive standalone actions

From the Google Apps Scripts Executions PDF:

hideReportTemplates: 15.22 sec

showReportTemplates: 15.376 sec

hideSystemSheetsNow: 11.357 sec

This confirms template visibility operations are not free. Deferring hides to one batch after template creation is correct, but the batch itself should skip already-hidden sheets and avoid unnecessary activation/show/hide cycles.

3. Architecture issues

3.1 Dashboard Quality section ordering is not centralized enough

The section map exists in RFF_DASHBOARD_QUALITY_SECTIONS, but many other functions still hard-code old section letters:

runDashboardQualityFull() hard-codes I/J/K/L/M/N/O.

updateDashboardQualitySignoffSection_() hard-codes Section J.

updateDashboardQualitySummarySection_() hard-codes Section K.

Summary row labels hard-code Section L/M/N/O.

The section registry should be the single authority, but it is not currently used consistently.

3.2 Removed tests are still represented as quality artifacts

The removed/retired tests still exist in dashboard storage functions and signoff/summary paths:

Test 7: Intentional Validation Failure

Test 9: Monthly Change Subheaders

Test 10: Dashboard Audit

Storage writers still save to those keys.

Signoff still displays Test 7 and Test 9.

Summary still displays Dashboard Audit.

3.3 Signoff and Summary run before the sections they summarize
Current workflow order:

Health

Signoff

Summary

Performance

Master List

Care Plan

Workflow

This guarantees Summary and Signoff can be stale or incomplete for sections that run afterward.

Correct order should be:

H — Health

I — Performance

J — Master List Health

K — Care Plan Sync

L — Workflow & Synchronization

M — Summary

N — Signoff

3.4 Section F should absorb Monthly Change Subheaders

User instruction says:

Monthly Change Subheaders - should be under section F

The current Section F only stages template validation rows from `validateReportTemplatesCore_()`.

Monthly Change subheader validation should be added to Section F as a subsection/row group, not stored under `RFF_TEST9_SUBHEADER_SHEET` and surfaced in Signoff.

3.5 v1.4.26.1 uses faster, narrower formatting patterns

The archive's final row-height lock is a simple bulk operation over all rows, using `setRowHeightsForced` or `setRowHeights` once.

The archive's Disenrolled path is also narrow:

copy template

set title/date/header rows

clear only output data range

write rows in chunks

set final row heights only on data rows

That fast pattern is visible here.

The current script's final output row-height lock loads dashboard config and applies several separate row-height runs plus data rows.

The current script also applies date/number formats column-by-column and then reapplies sample rows, with a `SpreadsheetApp.flush()` on final paths.

That design is safer for validation, but slower than the older archive's minimal formatting style.

4. Prioritized remediation plan

P0 — Correct Dashboard Quality section model

Update the single section registry to the new mapping:

Section	Title
A	Global Inputs Verification

- B Sheet Definitions Verification
- C Sheet Headers Verification
- D Column Definitions Verification
- E Sheet Behavior Verification
- F Template Structure & Validation
- G Date Format Integrity Validation
- H Framework Health Check
- I Performance Summary
- J Master List Health Check
- K Care Plan Sync Diagnostics
- L Workflow & Synchronization Verification
- M Summary
- N Signoff

Then update all hard-coded title/letter consumers:

RFF_DASHBOARD_QUALITY_SECTIONS

getDefaultDashboardQualityDetailHeader_

runDashboardQualityFull

runDashboardQualityTemplateAndFormatSections_

updateDashboardQualitySummarySection_

updateDashboardQualitySignoffSection_

buildFrameworkSummaryRows_

getTimingProcessNameForDashboardQualitySection_

any combined dashboard legacy section labels

P0 — Remove obsolete quality artifacts from Signoff/Summary
Remove from Signoff:

Intentional Validation Failure

standalone Monthly Change Subheaders

Dashboard Integrity if it depends on removed Dashboard Audit

Move Monthly Change Subheaders into Section F by appending it to template validation rows.

Remove Dashboard Audit from Summary entirely.

P0 — Fix Create Disenrolled undefined function

The report failure must be resolved before performance analysis for that workflow is meaningful.

Use the current defined helper:

`lockFinalOutputRowHeights_(sheet, "Disenrolled")`

or restore the v1.4.26.1 style helper if the intent is a fixed fast 25px lock. The archive's fast helper is simpler and likely faster.

P1 — Reorder Dashboard Quality Workflow

New workflow order should be:

Section H — Framework Health Check

Section I — Performance Summary

Section J — Master List Health Check

Section K — Care Plan Sync Diagnostics

Section L — Workflow & Synchronization Verification

Section M — Summary

Section N — Signoff

Summary and Signoff must be last.

P1 — Batch Dashboard Quality Workflow writes

Current Dashboard Quality Workflow runs sections individually through `runDashboardQualitySectionIfDue_()` and each section may save/style independently.

Instead:

Collect H–L rows in memory.

Build Summary M from the just-collected rows.

Build Signoff N from the same in-memory statuses.

Perform one `replaceDashboardQualitySectionsRows_()` call.

Perform one style pass for H–N.

Write timing once.

This should attack the 296–357 sec Dashboard Quality Workflow bottleneck.

P1 — Move Monthly Change Subheaders into Section F

Add Monthly Change subheader validation to the Section F staged write path after template validation rows. Current F/G batching already exists and should be extended there.

Do not keep it as standalone Test 9 or in Signoff.

P2 — Apply v1.4.26.1 fast formatting principles

From spec/v1.4.26.1_Production_Script_Archive.txt, the fastest approach is:

copy the verified template

write values once or in chunks

avoid reformatting full outputs after copy

apply final row heights with bulk row-height calls only

avoid column/row loops unless fallback is required

Archive examples:

fast final row-height lock.

fast Disenrolled output pattern.

Apply this to:

Raw Data output creation

Demo P processing

Disenrolled output

Master List output

Dashboard Quality section writes

P2 — Reduce template build formatting calls

Template build should keep the current governed model, but reduce expensive per-column/per-format operations:

Group number formats by identical format and contiguous runs instead of applying every column independently.

Avoid duplicate sample-row number-format writes except when validation requires repair.

Avoid final SpreadsheetApp.flush() unless a repair was actually performed.

Preserve deferred template hiding after full batch refresh.

Keep metadata-only refresh path strict and avoid hide/show/re-freeze there.

Current per-column date/number formatting is here.

P3 — Clean legacy combined dashboard functions

There are older combined-dashboard section labels that still reference old section structures and removed tests, including SECTION M - SIGNOFF and Test 7/Test 9 rows.

These should either be updated to the new A–N model or retired if Dashboard Quality Report is now the only governed output.